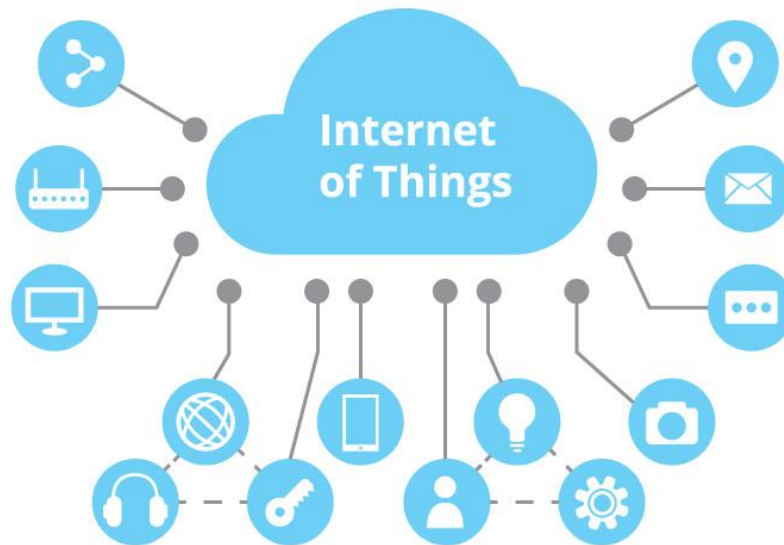


Internet of Things (IoT)



Chapter 5: NODE-RED

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1. Developing interface

Google

node red css



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Ui builder



Button



Dashboard node



Homey dashboard



iTube

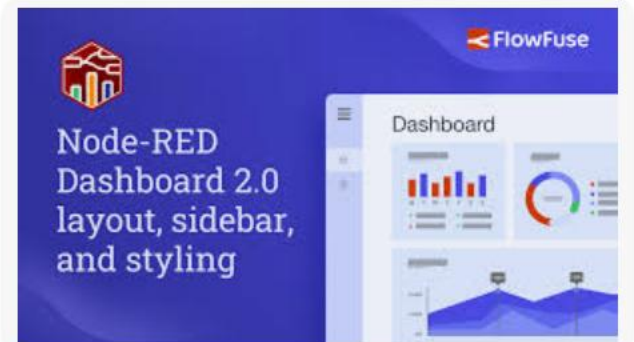
-RED Dashboard with some CSS hacks ...



Homey Community Forum

Node Red: A widget based dashboard

3



FlowFuse

Node-RED Dashboard 2.0 layout, sidebar ...

1. Developing interface

Node Red: A widget based dashboard working with Homey trough..

■ Apps



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The CSS code.

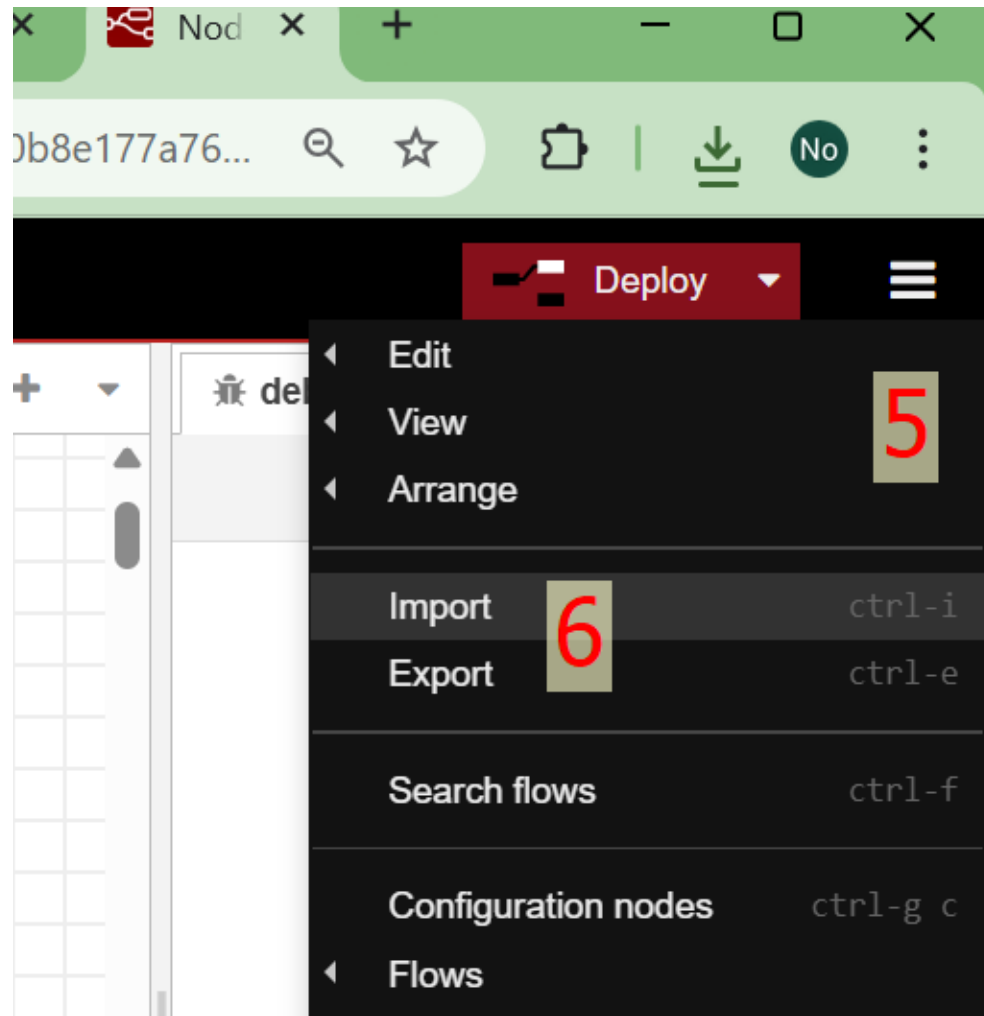
```
[{"id":"d81aa8bc.35b81","type":"ui_template","z":"d6fa9f2d.3d501","group":"339d0edc
```

4

With some modifications, you have the final Dashboard.



1. Developing interface



1. Developing interface

Import nodes

Clipboard

Paste flow json or  select a file to import

Local

Examples

```
!important;\n  border-radius: 10px !important;\n}\n\n.md-card.md-default-theme,\nmd-card {\n  border-radius: 10px;\n}\n\n.nr-dashboard-switch.ng-scope {\n  background-color: rgba(255,255,255,.2) !important;\n}\n\n.nr-dashboard-switch.ng-scope:hover {\n  background-color: rgba(255,255,255,.5) !important;\n}\n\n.nr-dashboard-theme .nr-dashboard-button .md-button {\n  background-color: rgba(255,255,255,.2);\n}\n\n.md-button {\n  border-radius: 10px;\n}\n\n.nr-dashboard-cardpanel layout-column {\n  background-color: rgba(255,255,255,0.2) !important;\n  background-color: transparent !important;\n  border-radius: 10px !important;\n}\n\n.nr-dashboard-theme ui-card-panel {\n  border: none;\n}\n\n.nr-dashboard-template {\n  overflow-y: visible;\n}\n\n.nr-dashboard-theme md-content md-card {\n  background-color: transparent !important;\n}\n\n.nr-dashboard-theme ui-card-panel p.nr-dashboard-cardtitle {\n  color: rgba(255, 255, 255, 0.5);\n}\n</style>\n\n<script>\ndocument.ontouchmove = function (e) {\n  e.preventDefault();\n}\n</script>","storeOutMessages":true,"fwdInMessages":true,"resendOnRefresh":false,"templateScope":"global","x":420,"y":160,"wires":[]], [{"id":"339d0edc.4ffe02","type":"ui_group","z":"","name":"Heure","tab":"a2490d0e.a872b","order":1,"disp":false,"width":"6","collapse":false}, {"id":"a2490d0e.a872b","type":"ui_tab","z":"","name":"Homey Dashboard","icon":"dashboard","order":1,"disabled":false,"hidden":false}]
```

Import to current flow new flow

Cancel

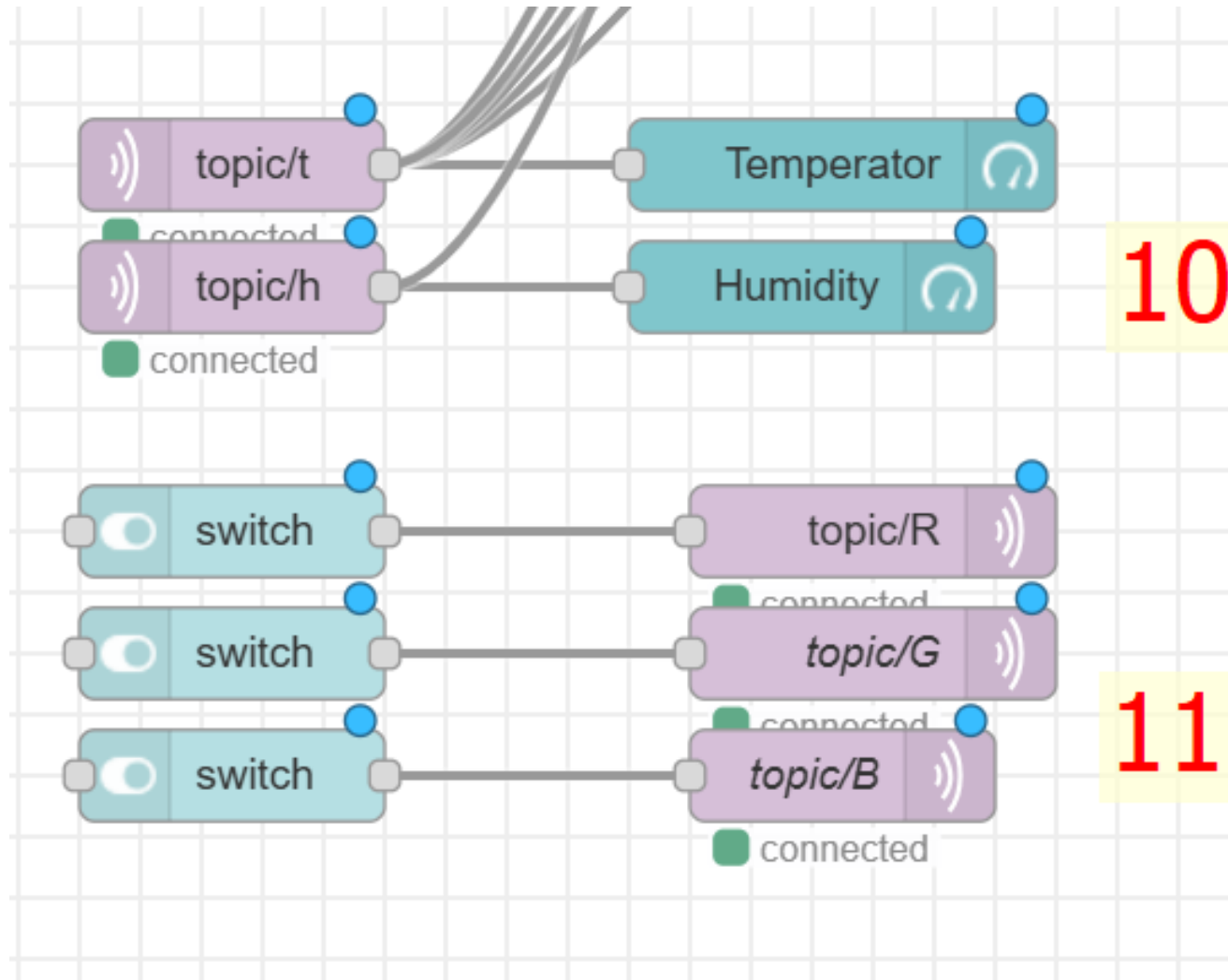
Import

 Node-Red Dashboard Custom CSS

9

8

1. Developing interface

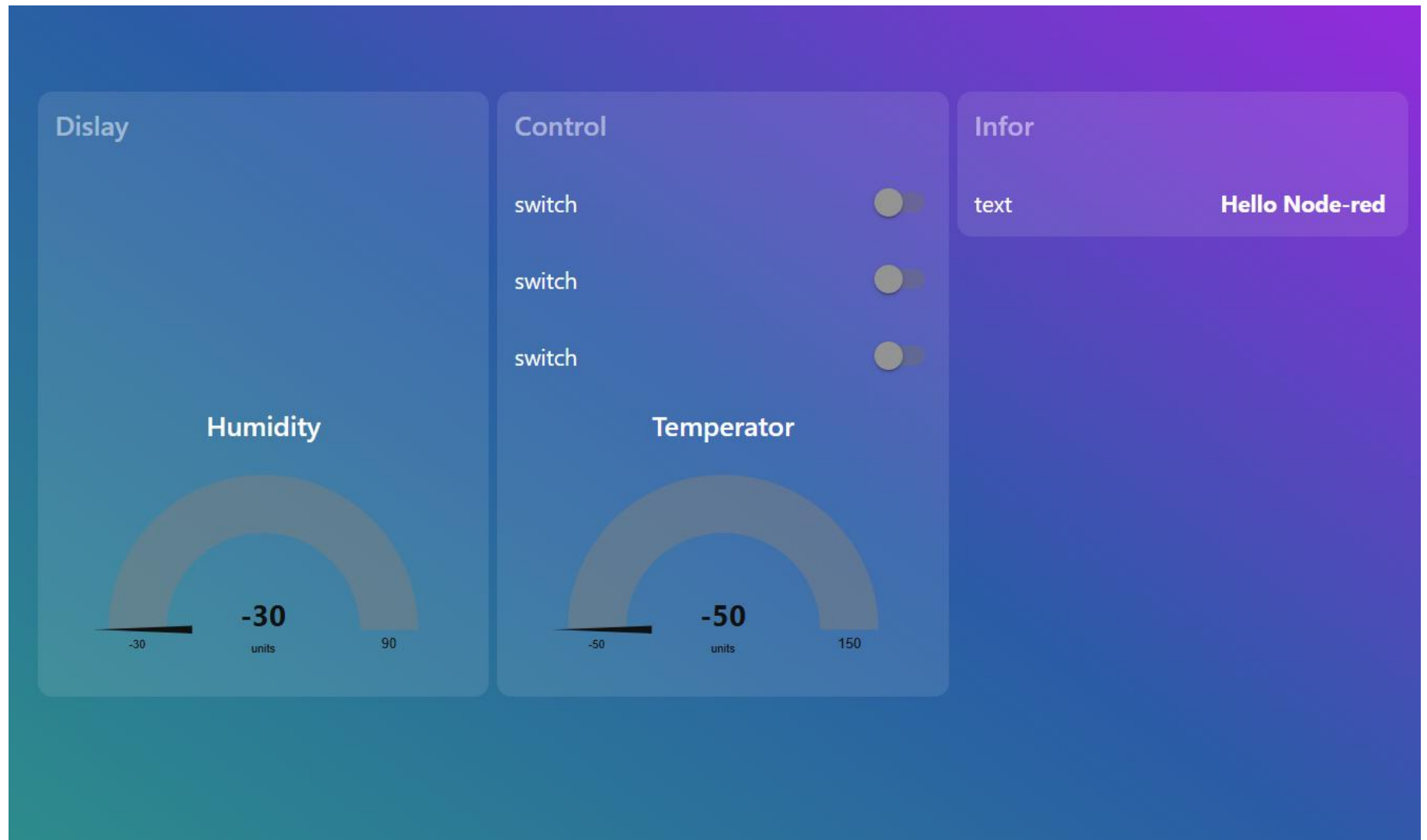


1. Developing interface

The image displays the Node-RED interface during development, featuring several key components:

- Inject Node (12):** A blue node with a right-pointing arrow icon and the label "inject". Below it is the "Edit Inject node" dialog, which includes a "Delete" button, "Cancel", and "Done" buttons. The "Properties" section shows a "Name" field and a "msg. payload" field set to "Hello Node-red".
- Text Node (13):** A teal node with the label "text" and the value "abc". Below it is the "Edit text node" dialog, which includes a "Delete" button, "Cancel", and "Done" buttons. The "Properties" section shows a "Group" dropdown menu set to "[Tab 1] Infor".
- Tabs & Links Panel:** A panel on the right side of the interface showing a tree view of tabs and links. The "Infor" tab is highlighted with a red box.
- Flow Diagram (14):** A flow diagram on a grid background showing a sequence of nodes. The first node is a blue "Hello Node-red" node, followed by a teal "text" node with the value "abc". The "text" node is highlighted with an orange border.

1. Developing interface



2. Saving data

▼ storage



1



2

Edit write file node

Delete

3

Cancel

Done

⚙ Properties

📄 Filename

▼ path C:\Users\tanno\Desktop\data.txt

🔄 Action

append to file ▼

☒ Add newline (\n) to each payload?

2. Saving data

function

function

4

topic/t

connected

function 1

5

C:\Users\tanno\Desktop\data.txt

debug 3

Edit function node

Delete

Cancel

Done

Properties

Name

function 1

Setup

On Start

On Message

On Stop

```
1 msg.payload = getCurrentDateTime() + " received: " + msg.payload + " C\n";
2 return msg;
3 function getCurrentDateTime() {
4     const now = new Date();
5
6     const dd = String(now.getDate()).padStart(2, '0');
7     const mm = String(now.getMonth() + 1).padStart(2, '0'); // Tháng bắt đầu từ 0
8     const yyyy = now.getFullYear();
9
10    const hh = String(now.getHours()).padStart(2, '0');
11    const min = String(now.getMinutes()).padStart(2, '0');
12    const ss = String(now.getSeconds()).padStart(2, '0');
13
14    return `${dd}/${mm}/${yyyy} ${hh}:${min}:${ss}`;
15 }
```

6

2. Saving data

▼ common

⇒ inject **7**



Edit write file node

Delete Cancel Done

⚙ Properties

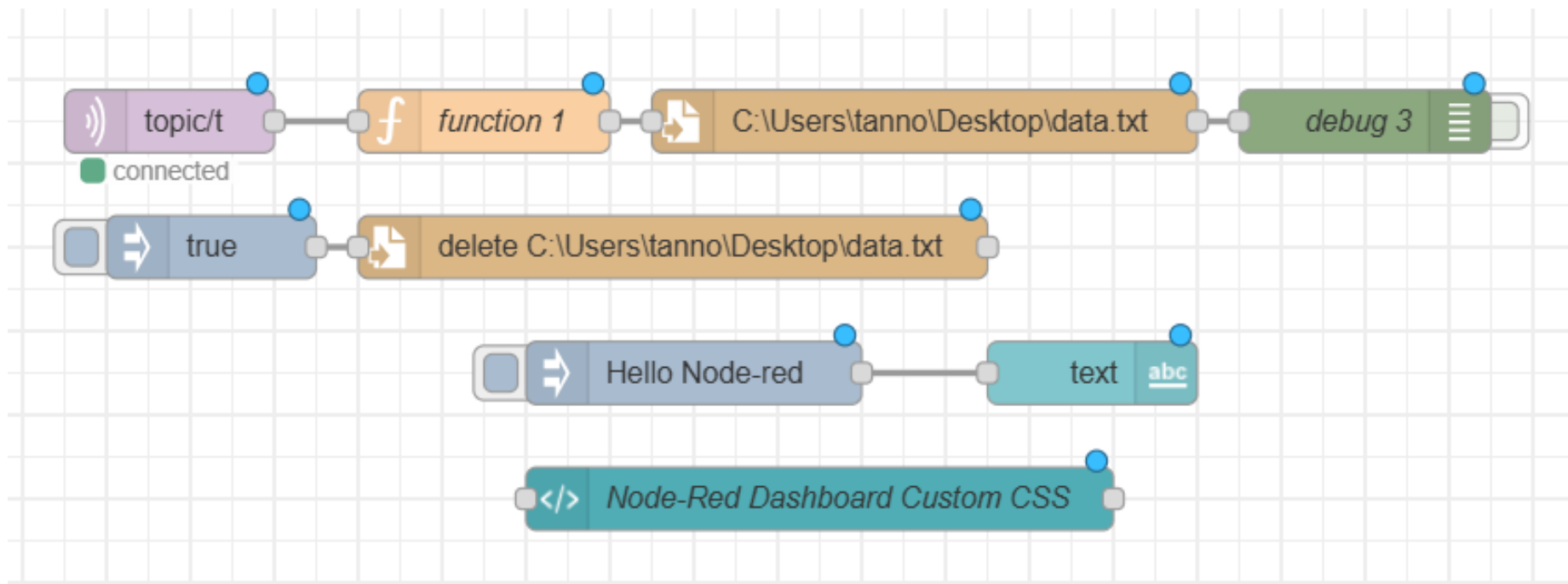
📁 Filename ▼ path C:\Users\tanno\Desktop\data.txt

🔄 Action delete file ▼

🏷 Name **9**

Tip: The filename should be an absolute path, otherwise it will be relative to the working directory of the Node-RED process.

2. Saving data



3. Code

```
1  #include <Arduino.h>
2  #include <Wifi.h>
3  #include <PubSubClient.h>
4
5  const char *ssid = "ESP32";
6  const char *password = "88888888";
7
8  const char *mqtt_broker = "broker.emqx.io";
9  const char *mqtt_user = "emqx";
10 const char *mqtt_pass = "public";
11 const int mqtt_port = 1883;
12
13 WiFiClient espClient;
14 PubSubClient client(espClient);
```

3. Code

```
26   Serial.begin(9600);
27   WiFi.begin(ssid, password);
28   while (WiFi.status() != WL_CONNECTED)
29   {
30       delay(100);
31       Serial.print(".");
32   }
33   Serial.println("\nWiFi connected");
```

3. Code

```
36 client.setServer(mqtt_broker, mqtt_port);
37 client.setCallback(callback);
38 while (!client.connected())
39 {
40     String clientId = "esp32-" + String(WiFi.macAddress());
41     if (client.connect(clientId.c_str(), mqtt_user, mqtt_pass))
42     {
43         Serial.println("MQTT connected");
44     }
45     else
46     {
47         Serial.print("MQTT failed, state: ");
48         Serial.println(client.state());
49         delay(2000);
50     }
51 }
```

3. Code

```
20 unsigned long lastSend = 0; // Biến dùng millis
21 const unsigned long interval = 1000; //3 giây
22 void loop()
23 {
24     client.loop();
25     unsigned long now = millis();
26     if (now - lastSend >= interval)
27     {
28         lastSend = now;
29         float t = random(10, 50); // Simulate temperature data
30         float h = random(-30, 60); // Simulate humidity data
31         if (!isnan(t))
32             client.publish("topic/t", String(t, 2).c_str());
33         if (!isnan(h))
34             client.publish("topic/h", String(h, 2).c_str());
35         Serial.print(h);
36         Serial.print(" ");
37         Serial.println(t);
38     }
39 }
```

3. Code

```
client.subscribe("topic/R");
client.subscribe("topic/G");
client.subscribe("topic/B");

57 void callback(char *topic, byte *payload, unsigned int length)
58 {
59     String msg;
60     for (unsigned int i = 0; i < length; i++)
61         msg += (char)payload[i];
62     Serial.print(msg);
63     bool on = (msg == "true");
64     if (strcmp(topic, "topic/R") == 0)
65         digitalWrite(led_blue, on);
66     else if (strcmp(topic, "topic/G") == 0)
67         digitalWrite(led_red, on);
68     else if (strcmp(topic, "topic/B") == 0)
69         digitalWrite(led_green, on);
70 }
```

Thank for attention